

MOHAMMAD QATAADAH ZAKIR

Software Engineer

Riyadh, Saudi Arabia | mqzakir@outlook.com | <https://linkedin.com/in/mqzakir> | <https://mqzakir.com> | <https://github.com/mqzakir>

PROFESSIONAL SUMMARY

Software Engineer experienced in designing and deploying scalable backend systems, cloud-native applications, and modern software solutions using Python, FastAPI, Docker, PostgreSQL, and AWS. Skilled in building REST APIs, distributed services, automation pipelines, and high-performance backend infrastructure, with additional expertise in Artificial Intelligence, Machine Learning, and Large Language Models.

WORK EXPERIENCE

AI Software Engineer | Optimal Communication Co.

July 2025 – Present

- Engineered and deployed AI-powered payment automation solutions for POS terminals, card payment systems, and internal business workflows, developing scalable backend services with Python, FastAPI, REST APIs, SQL, and Git that reduced manual processing by 65%, improved transaction accuracy by 30%, and supported 15+ production features.
- Designed, built, and maintained full-stack software applications from requirements through deployment, integrating payment gateways, third-party APIs, authentication systems, databases, and business automation tools, decreasing workflow completion by 45% while improving system reliability and maintainability through comprehensive testing, debugging, and code reviews.
- Collaborated closely with product managers, QA engineers, and stakeholders to deliver 10+ software enhancements and automation initiatives, optimizing application performance, resolving critical production issues, and implementing CI/CD and version control best practices that reduced deployment times by 40% and improved release stability.

Software Engineer – Intern | Modern Arc Group

May 2024 – August 2024

- Assisted in the development and enhancement of web-based business applications using modern software engineering practices, contributing to frontend and backend development, REST API integration, SQL database management, debugging, testing, and documentation, while helping deliver 5+ client and internal software solutions within project deadlines.
- Worked alongside senior software engineers to analyze business requirements, implement new features, troubleshoot complex software defects, and participate in peer code reviews, successfully resolving 70+ reported issues and improving application performance, maintainability, and user experience across multiple development cycles.
- Contributed to software deployment, database optimization, and technical documentation while gaining experience with Git, Agile/Scrum methodologies, SDLC, version control, and cross-functional collaboration, reducing onboarding time for new developers by 30% and supporting faster feature delivery throughout the internship.

Contract Software Engineer

June 2022 – January 2024

EDUCATION

Software Engineering, B.Eng. (Hons)

University of Greater Manchester (formerly University of Bolton) | 2025

Computer Engineering, B.Eng.

Memorial University of Newfoundland (MUN) | 2022 (Credit transfer)

PROJECTS

QEMI – Intelligent Software Engineering Platform

Engineered a cross-platform desktop software engineering platform using React, Electron, Python, and FastAPI, delivering real-time system monitoring, intelligent debugging, and AI-assisted code analysis through a modular client-server architecture. Designed as a production-ready MVP integrating modern frontend engineering, backend services, and embedded LLM capabilities. The platform leverages WebSockets, RESTful APIs, and local AI inference to provide contextual diagnostics, runtime insights, and intelligent recommendations while maintaining a responsive, low-latency user experience.

FeelTok – Social Networking Platform

Designed and developed a full-stack cross-platform social networking application focused on meaningful user engagement, featuring secure authentication, real-time content management, and scalable cloud-backed infrastructure. Built using React Native, Node.js, Supabase, and PostgreSQL following Agile software development practices. The application was architected with a scalable backend, secure cloud storage, and extensible data models to support future social features, real-time interactions, and seamless cross-device synchronization.

SKILLS

Programming languages – Python, SQL, Javascript, Typescript, Bash, C#, Java

Frontend – React, React Native, Expo, Tailwind CSS, HTML5, CSS3, Avalonia, WPF

Backend – FastAPI, Flask, Django, ASP.NET, Node.js, Express.js, REST APIs, WebSockets, Pydantic

Databases – PostgreSQL, MongoDB, Supabase, SQLite

Cloud & Infrastructure – AWS, Docker, Kubernetes, Git, CI/CD, Linux

AI & Machine Learning – OpenAI API, Claude, Cursor, Embedding APIs, Vector Databases, LangChain, Together AI, Zapier, LangGraph, PyTorch, TensorFlow, Scikit-learn, XGBoost, ONNX

Soft skills – Agile, Cross-functional Collaboration, Communication, Client Collaboration, Documentation, Adaptability, Time Management, Emotional Intelligence

CERTIFICATIONS

- University of Minnesota – Software Development Lifecycle
- Dartmouth University – C Programming with Linux
- Duke University – Java Programming and Software Engineering Fundamentals
- EDUCBA – AI with Python: Foundations to Projects Specialization
- DeepLearnin.AI – Deep Learning Specialization
- Stanford Online – Machine Learning Specialization
- HarvardX - Computer Science for Python Programming
- IBM – AI Chatbots without Programming
- IBM – Python basics for Data Science